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/ Load-Dump Protection for 24V Automotive Applications

Load-Dump Protection for 24V Automotive Applications

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Abstract

As self-driving vehicles, passenger safety, and vehicle-to-vehicle communication evolve, one of the key challenges for automotive manufacturers is to protect the electronic system from high-energy transient pulses. The high-energy transient pulse, or load-dump pulse, is one of the most destructive pulses described in ISO 16750-2 under test A and test B (formerly 5a and 5b in ISO 7637-2). In this application note, we show how to protect your downstream subsystem from a load-dump pulse in 24V systems using the [MAX16126](/en/products/max16126.html) (</en/products/max16126.html>), [MAX16127](/en/products/max16127.html) (</en/products/max16127.html>), and external TVS diodes. Other protection features of the MAX16126/MAX16127 include undervoltage/overvoltage, overcurrent, reverse voltage/current, and overtemperature protections.

Introduction

Semiconductor components are transforming the automotive industry by adding a wide-range of innovations in self-driving vehicles, passenger safety, and vehicle-to-vehicle communication. As passenger safety is the primary objective of advanced driver-assistance systems (ADAS), increasing numbers of auto manufacturers are adding more safety and reliability ICs to their automotive systems.

A key component of system safety and reliability includes a robust power management and subsystems protection design, which prevents downstream damage from high-energy transient pulses. In this application note, we show how to protect your downstream subsystem from a load-dump pulse in a 24V systems as per the ISO 16750-2 specification for test A (without centralized load-dump suppression) and test B (with centralized load-dump suppression) pulses. These test pulses were previously described in ISO 7637-2 as 5a and 5b, respectively.

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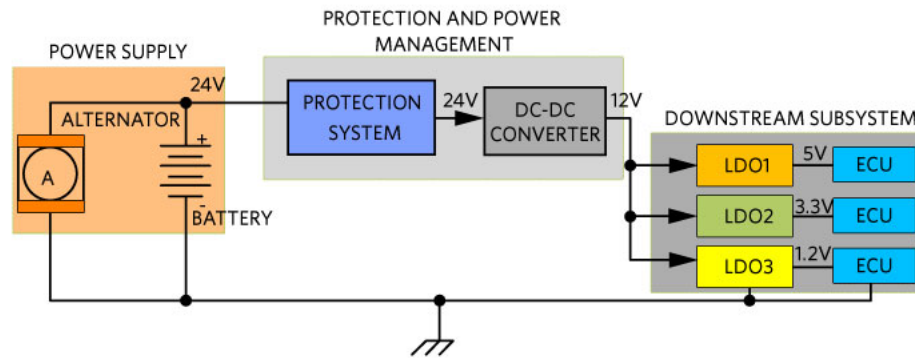


Figure 1. Power architecture inside an automotive system.

To understand how to protect a subsystem from a load-dump pulse, we must first understand the architecture inside an automotive system.

Automotive vehicles can be divided into two classes: lightweight and heavyweight vehicles. Lightweight vehicles like cars typically use a 12V battery, whereas heavyweight vehicles like trucks, cranes, and tractors typically use 24V batteries. A common power architecture inside an automotive system consists of a battery, alternator, protection system block, DC-DC converters, and downstream subsystems that include LDOs and different functional ECUs, as shown in Figure 1.

Within the automotive system, the downstream subsystem contains the most sensitive components, so the challenge for system designer is to reliably regulate the automotive battery voltage to power these sensors and processors within the ECUs.

Now we will look at what causes a load dump and how it affects the subsystems in this design.

Load Dump

In automotive systems, a load-dump transient occurs when an alternator is delivering current to a battery and the battery is abruptly removed. This can happen when a discharged battery loses connectivity while the alternator is generating charging current and other loads are connected to the alternator circuit, as shown in Figure 2.

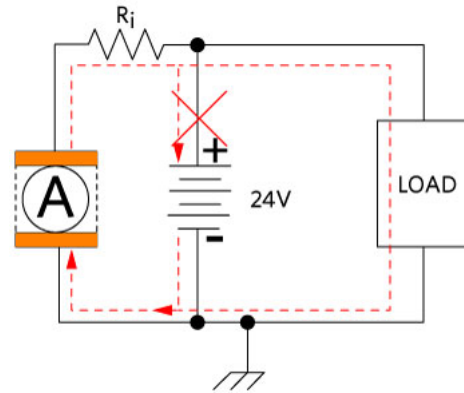


Figure 2. Load-dump generation.

To understand how the load dump affects the downstream subsystem, we need to understand the components of the load dump.

The load-dump event has two components. First is the pulse amplitude (V_p), which depends on the alternator speed and the level of the alternator's field excitation at the time when the battery is disconnected. Second is the pulse duration (t_d), which depends on the time constant of the field excitation circuit and V_p , as shown in Figure 3. The V_p can be as high as 202V and may take up to 350ms to decay. The amount of energy dumped to the load depends on the internal resistance (R_i) of the source, as shown in Figure 2.

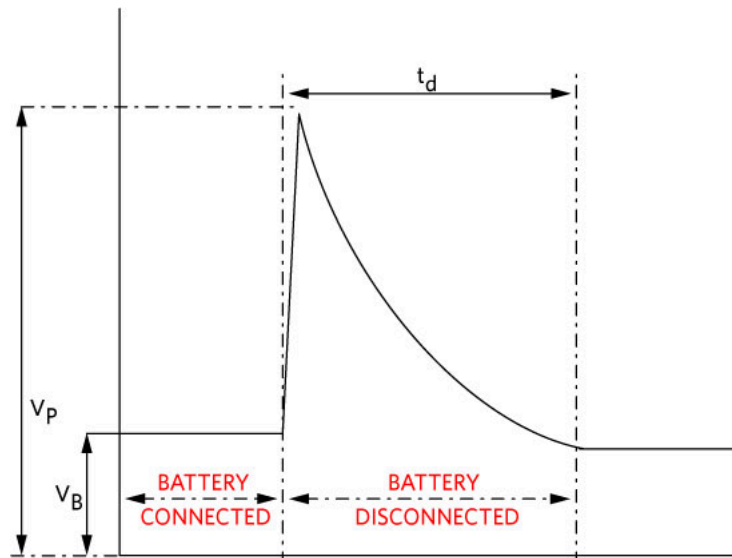


Figure 3. Output voltage of an alternator during a load-dump event showing nominal battery voltage (V_B).

ISO Standard for Load-Dump Pulse

Standards often evolve to meet the design challenges of today's technology. Prior to 2011, load dump was initially defined as 5a and 5b pulses in the ISO 7637 standard entitled "Road vehicles—Electrical disturbances from conduction and coupling" and was considered an electromagnetic compatibility (EMC) specification, which included transients related to supply quality.

In 2011, the specifications related to power-supply quality but not EMC were moved to the ISO 16750-2 standard, "Road vehicles—Environmental conditions and testing for electrical and electronic equipment, Part 2: Electrical loads," placing load-dump pulse under test A and test B pulses.

As safety and reliability became the priority for all automotive systems, automotive manufacturers naturally migrated away from their own load-dump standards and adopted the ISO standard, utilizing the load-dump test A pulse specified under ISO 16750-2.

Load-Dump Protection

There are a variety of ways to protect automotive systems from load-dump events. In this application note, we cover the three most effective methods of load-dump protection along with the pros and cons of each type of method.

In the first method, we isolate the load from the transient source. In our second method, we clamp the load-dump pulse to a safe voltage (abs max rating) of the front-end DC-DC converter. And in the third method, we use a hybrid solution that is a combination of both the first and the second methods.

Method 1: Isolate the Load from the Transient Source Using the MAX16126/MAX16127

In this approach, we isolate the load-dump source from the downstream load by using a power MOSFET as a switch and use the MAX16126/MAX16127 protection IC to control the MOSFET, as shown in Figure 4. The MAX16126/MAX16127 monitors the input rail, and if the input rail exceeds the overvoltage threshold, it turns off the MOSFET to isolate the transient source from the load.

The MAX16126/MAX16127's wide operating range allows a system designer to set the overvoltage and undervoltage thresholds between 3V to 30V using a resistor network. The wide protection range of -36V to +90V provides a reliable solution to prevent load-dump and battery-reversal conditions. This makes the MAX16126/MAX16127 the ideal solution for automotive applications.

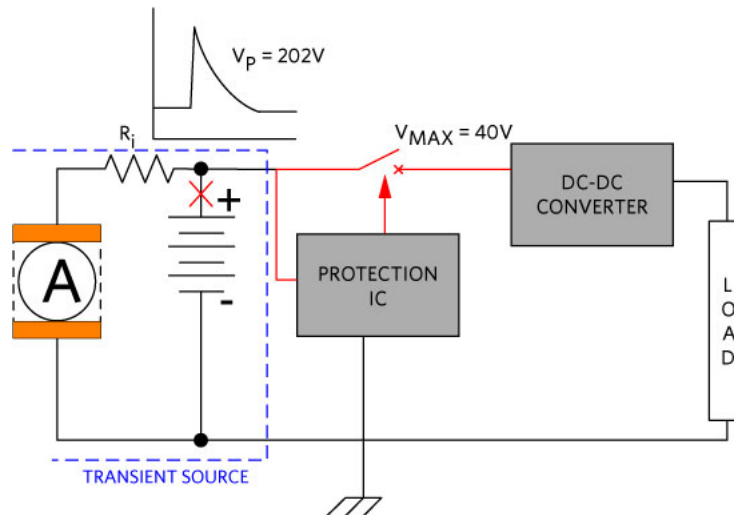


Figure 4. Isolation of the load-dump source from the downstream load.

To increase the protection range of the MAX16126, designers can use a zener diode at the positive supply input-voltage pin (IN) with a current-limiting resistor (R_p) in series, as shown in Figure 5. During a load-dump event, if IN overshoots by more than +90V, the zener diode will clamp the MAX16126 at +90V and R_p will limit the current through the zener diode.

In this method, the selection of a series MOSFET is very important, as the complete load-dump voltage will appear across the drain and source of MOSFET. The MOSFET must be chosen with a maximum V_{DS} of more than 1.2 times the peak load-dump pulse. Because a MOSFET with higher V_{DS} has more $R_{DS(ON)}$, it will decrease the overall efficiency of the system during normal operation.

The benefit of this method is that it can protect from a variety of events including load dump, cold crank, and voltage reversal. Setting the overvoltage and undervoltage thresholds is as easy as adding the appropriate resistor.

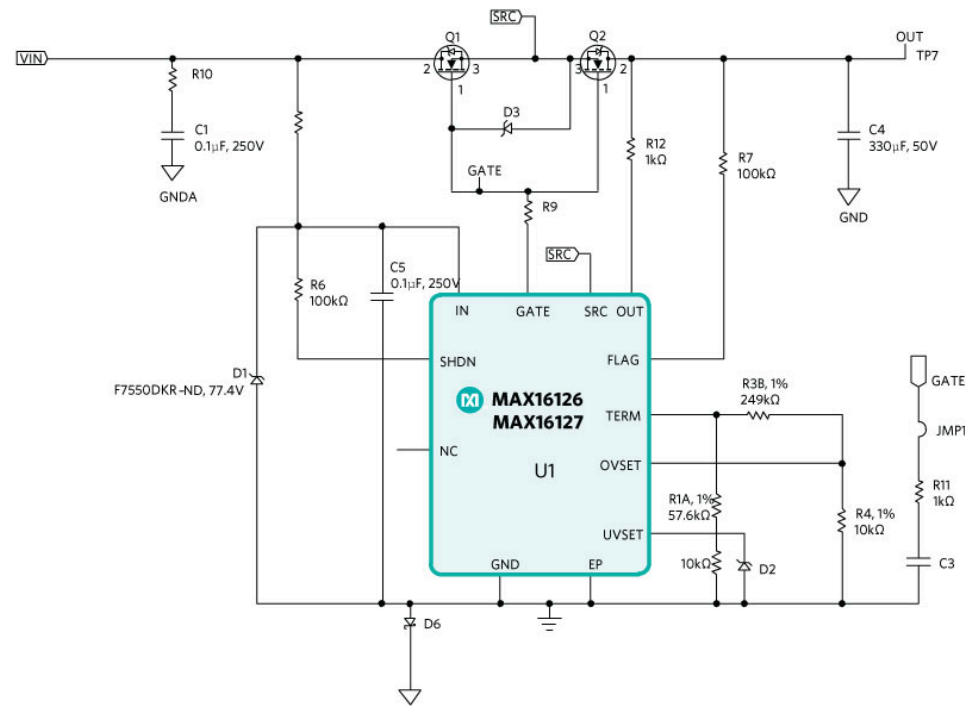


Figure 5. Increase the protection range of the MAX16126/MAX16127.

Method 2: Clamp the Load-Dump Pulse Using a TVS Diode

In this approach, we clamp the transient pulse to the safe voltage (abs max voltage) of the DC-DC converter by using a transient voltage suppressor (TVS) diode, as shown in Figure 6. The TVS diode clamps the transient pulse and absorbs the maximum energy of the pulse. Note that the internal resistance (R_i) of the alternator in the ISO 16750-2 test A pulse specification is between 10 and 80. This limits the maximum energy delivered to the TVS diode. It is very important to select the appropriate TVS diode, as the energy of the pulse depends on the clamping voltage, pulse duration, and R_i of the load-dump source.

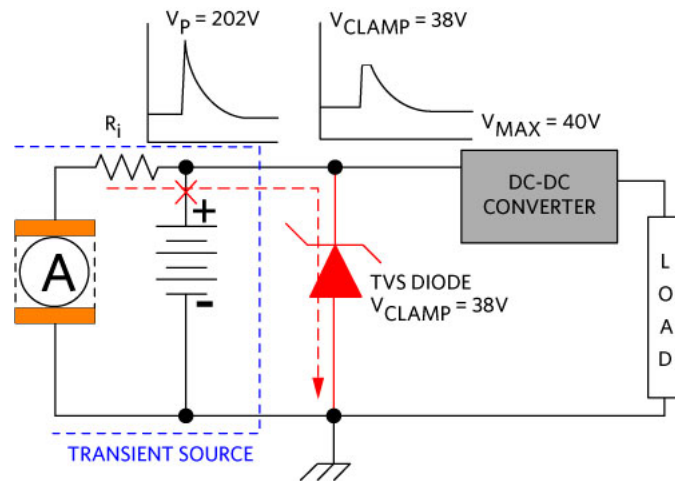


Figure 6. Clamp the load-dump pulse to a safe operation voltage.

To demonstrate this, we calculate the peak power (P_{PK}) for a test A pulse where $V_{Sa} = 151V$, $R_i = 10$, $t_d = 100ms$, and $V_{CLAMP} = 38V$.

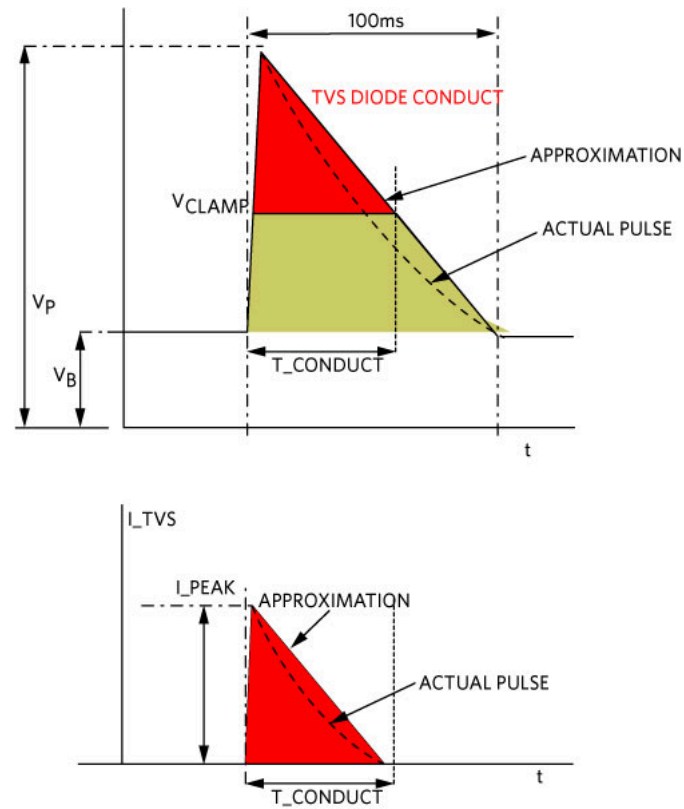


Figure 7. Approximate area of a pulse absorbed by a TVS diode during a load dump.

$$I_{PEAK} = \frac{V_S - V_{CLAMP}}{R_i}$$

$$= \frac{151 - 38}{1} = 113A$$

$$P_{PK} = I_{PEAK} \times V_{CLAMP}$$

$$= 38 \times 113 = 4.294kW$$

To calculate approximately the time that the TVS diode will conduct, as well as the energy absorbed by the TVS diode, we consider the pulse shape as triangular as shown in Figure 7 and calculate the TVS diode conduction time (T_CONDUCT) as follows:

$$\begin{aligned}\text{Slope of pulse} &= \frac{V_S}{t_d} \\ &= \frac{151V}{100ms}\end{aligned}$$

$$\begin{aligned}T_{\text{CONDUCT}} &= \frac{V_S - V_{\text{CLAMP}}}{\text{Slope of pulse}} \\ &= \frac{(151 - 38) \times 100ms}{151} \\ &= 74.83ms\end{aligned}$$

Energy (E) absorbed by the TVS diode can be calculated using the following equation:

$$E = \int_0^{T_{\text{CONDUCT}}} V \times I dt$$

Here, the voltage that is clamped by the TVS diode (V_{CLAMP}) is 38V, and I is a triangle pulse having $I_{\text{PEAK}} = 113A$ and a duration of 74.83ms.

$$\begin{aligned}E &= V_{\text{CLAMP}} \times \int_0^{T_{\text{CONDUCT}}} I dt \\ &= \frac{V_{\text{CLAMP}} \times I_{\text{PEAK}} \times T_{\text{CONDUCT}}}{2} \\ &= 160J\end{aligned}$$

The challenge with this approach is that it is difficult to choose a single TVS diode with the peak power rating and pulse duration rating capable of absorbing the maximum energy of the pulse. For example, Figure 8 shows a typical curve for the peak power rating vs. pulse duration. As the peak pulse power increases, the capability of the TVS diode to sustain the pulse duration will decrease. This makes it difficult to find the ideal TVS diode for a high-energy load-dump pulse.

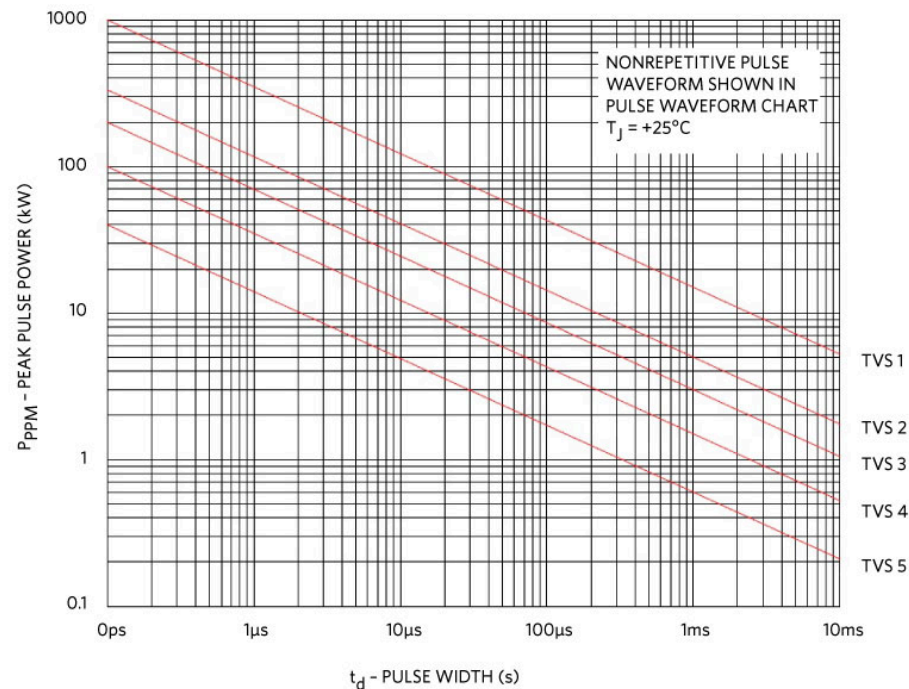


Figure 8. Peak pulse vs. maximum pulse duration of a typical TVS diode.

Note: The Figure 8 curve shows the general performance of different TVS diodes where TVS 1 = 15kW, TVS 2 = 5kW, TVS 3 = 3kW, TVS 4 = 600W, and TVS 5 = 400W.

The ISO 16750-2 specification requires the system to withstand 10 consecutive pulses with a 1-minute interval, and a TVS diode will degrade after each load-dump event. This makes it very difficult to design a system to protect against a load-dump pulse using a TVS diode alone.

Method 3: Isolate the Load from the Transient Source Using the MAX16126/MAX16127 and a TVS Diode

In this approach, we combine methods 1 and 2: a series switch and clamping the input voltage. In the first method, we were limited by the protection range of the IC and used a high-voltage MOSFET, and in the second method we were limited by the peak power

dissipation of the TVS diode. By using both a TVS diode and a protection IC, we can limit the peak power dissipation by increasing the clamping voltage of the TVS diode and isolating the load using a protection IC, which protects it from the load dump as shown in Figure 9.

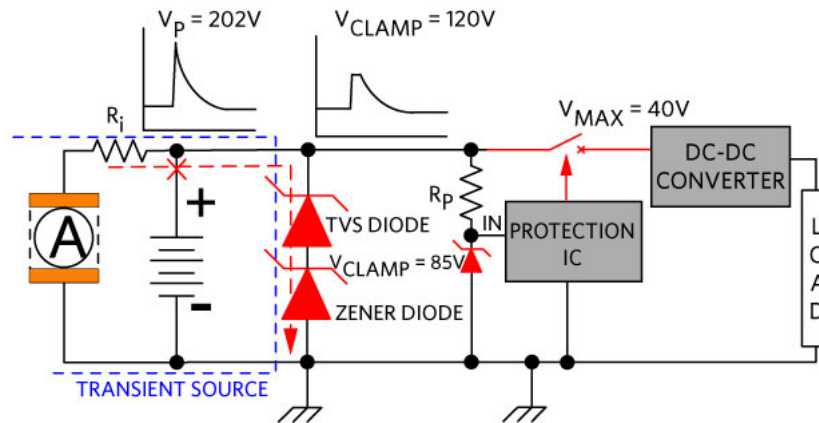


Figure 9. Load-dump protection using a TVS diode and protection IC.

To demonstrate this, we calculate the peak power of an ISO 16750-2 test A pulse where $V_{Sa} = 151V$, $R_i = 10\Omega$, $t_d = 100ms$, and $V_{CLAMP} = 120V$.

$$I_{PEAK} = \frac{V_S - V_{CLAMP}}{R_i}$$

$$= \frac{150 - 120}{1} = 30A$$

$$P_{PK} = I_{PEAK} \times V_{CLAMP}$$

$$= 120 \times 30 = 3.6kW$$

$$P_{PK \text{ per TVS}} = \frac{3.6}{2} = 1.8kW$$

We consider the same pulse as shown in Figure 7 to calculate $T_{CONDUCT}$:

$$T_{CONDUCT} = \frac{V_S - V_{CLAMP}}{\text{Slope of pulse}}$$

$$= \frac{(151 - 120) \times 100ms}{151}$$

$$= 19.86ms$$

Here, V_{CLAMP} is a voltage that is clamped by a 120V TVS diode where I is a triangle pulse with $I_{PEAK} = 30A$ and a duration of 19.86ms.

$$\begin{aligned}
 E &= V_{CLAMP} \times \int_0^{T_{CONDUCT}} I dt \\
 &= \frac{V_{CLAMP} \times I_{PEAK} \times T_{CONDUCT}}{2} \\
 &= 35.74J
 \end{aligned}$$

The peak power dissipation and pulse duration for the TVS diode is lower at 1.8kW and 19.86ms, making it easier to find the correct TVS diode. To increase the protection limit of the protection IC, we can also use a zener diode with a current-limiting resistor, as in method 1. The advantage of combining both methods allows designers to have the flexibility to use a low-voltage MOSFET.

To calculate the peak power dissipation across the zener diode (I_{ZENER_PK}), we use the same equations that were used previously. We assume after the TVS diode: $V_{S_TVS} = 120V$, $V_{ZENER_CLAMP} = 85V$, and $R_p = 1k\Omega$.

$$\begin{aligned}
 I_{ZENER_PK} &= \frac{V_{S_TVS} - V_{ZENER_CLAMP}}{R_p} \\
 &= \frac{120 - 85}{1000} = 35mA
 \end{aligned}$$

$$\begin{aligned}
 P_{PK} &= I_{ZENER_PK} \times V_{ZENER_CLAMP} \\
 &= 85 \times 35 = 2.975W
 \end{aligned}$$

$$\text{Power dissipation across } R_p = (V_S - V_{S_TVS}) \times I_{ZENER_PK}$$

$$P_{PK} \text{ dissipation across } R_p = (120 - 85) \times 0.035 = 1.225W$$

Based on the maximum power rating of the device calculated above, we chose a zener diode and an R_p for our application. An important consideration is the limitation of R_p , because as the value of R_p increases, the time response to detect the overvoltage and undervoltage faults will also increase. This is due to the time constant introduced by R_p and $C1$, as shown in Figure 10. To select the overvoltage and undervoltage thresholds, use the appropriate $R1$, $R2$, and $R3$ resistor values outlined in the MAX16126 data sheet.

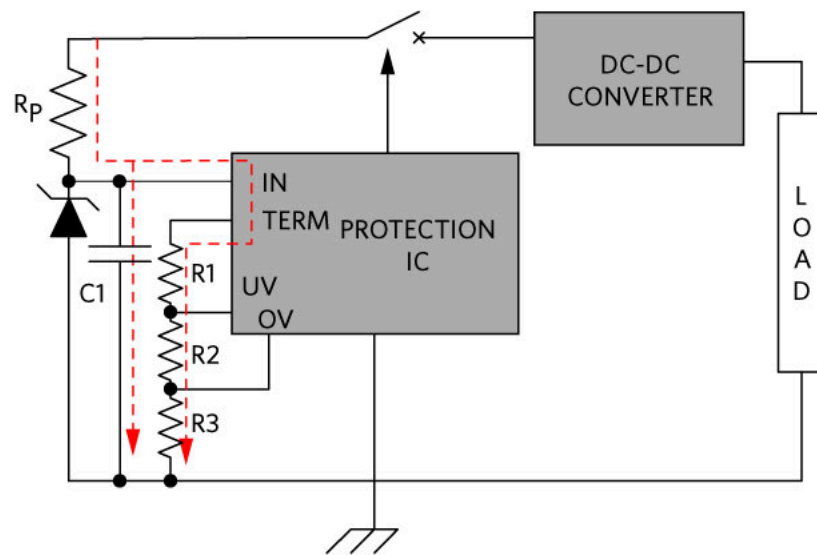


Figure 10. Time constant introduced by R_p and $C1$.

Testing and Lab Results Using Method 3

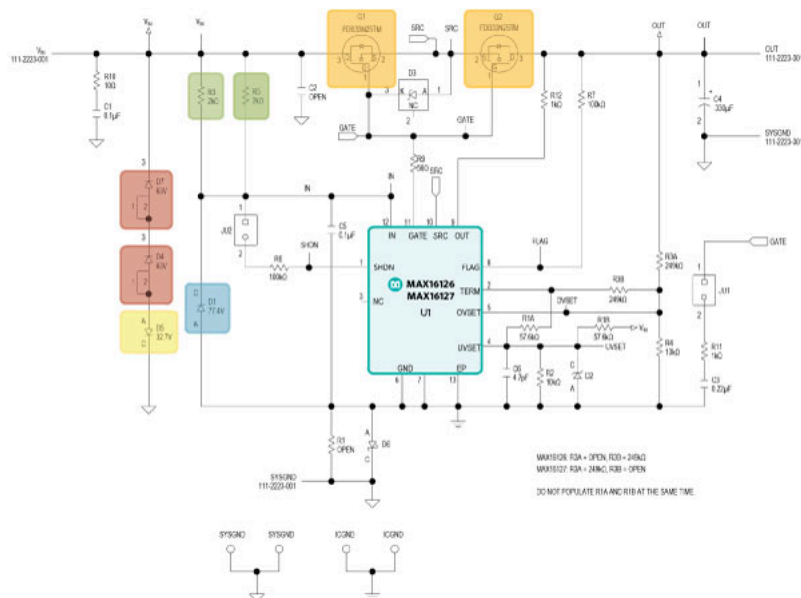
ISO 16750-2 specifies that a peak pulse for a 24V load-dump system can vary from 151V to 202V using different R_i values from 10 to 80. To test the corner case, we used different voltage levels, internal resistance, and pulse duration to demonstrate the robustness of our reference design board shown in Figure 11 and its ability to pass the toughest portions of the ISO specifications. TVS diodes D7 and D4 in Figure 12 are used to clamp the high-energy load-dump pulse to 120V.

- MOSFETs Q1 and Q2 are chosen with a 150V rating so that they can handle the 120V clamped load-dump pulse.
- Diode D1 is used to clamp the 120V pulse to +85V so that the IC does not exceed the maximum protection range.
- Current-limiting resistor R_P (R_3) is chosen at 1k Ω so that it can limit the power to diode D1.
- Diode D5 is used to handle the ISO 7637 negative pulse. D5 clamps the negative pulse to -32V to so that the IC cannot exceed its maximum protection voltage of -36V. Diode D6 is redundant and can be removed if we are using diode D5.



Figure 11. Load-dump reference board.

Figure 12 shows the schematic used to make the reference design.



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Figure 12. Schematic of a load-dump circuit using the MAX16126.

Figure 13 shows the result of the load-dump pulse of $V_S = 202V$, $R_i = 80\Omega$, and $t_d = 400ms$ applied to the MAX16126 reference board. In this setup, we use a $330\mu F$ output capacitor and $1A$ load. The overvoltage is set to $28V$. When a $202V$ pulse is applied to system diode D7, as seen in Figure 13, D4 clamps the load-dump pulse to $111V$ (blue trace). The MAX16126 detects an overvoltage fault when IN crosses $28V$, resulting in the turnoff of the MOSFET gate (pink trace); output is limited to $27V$ (yellow trace). To show the robustness of our $24V$ solution, we sent our reference board to Wipro® lab. The results from Wipro lab shows that our solution successfully passed the pulse test from $151V$ to $202V$ with an R_i range from 10Ω to 80Ω , thus meeting the ISO 16750-2 and ISO 7637-2 specifications.

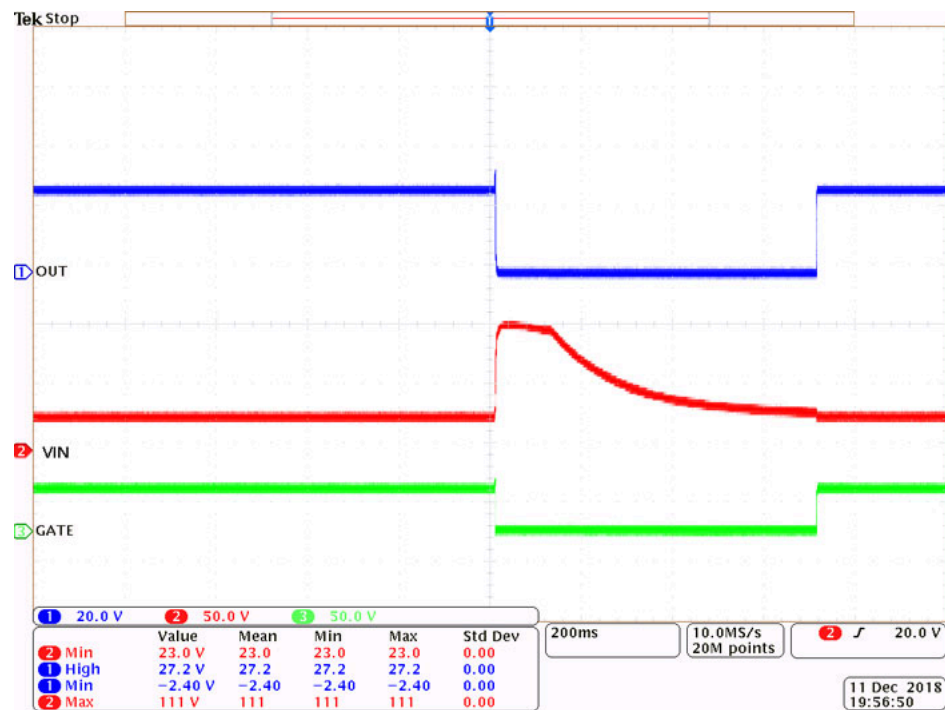


Figure 13. Load-dump pulse applied to the MAX16126.

Table 1 summarizes all three methods and their associated features.

Table 1.

Feature	Method 1	Method 2	Method 3
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Load-Dump Pulse	Up to +90V	Up to +202V	Up to +202V
Undervoltage Protection	Yes	No	Yes
Battery Reversal	Yes	No	Yes
Energy Absorbed by TVS Diode	N/A	160J	35.74J
Solution Size	Small	Bulky*	Small

*Because of the high energy absorbed by the TVS diode, we need to put a heatsink near the TVS diode.

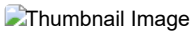
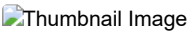
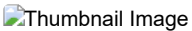
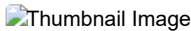
Summary

Automobile manufacturers are continually adding more semiconductor components to improve passenger safety. These systems need to operate safely and reliably, and the best way to accomplish this is with a robust power management and subsystem protection design that prevents downstream damage from high-energy transient pulses.

The alternator is the main cause of transient pulses when the battery loses connectivity, and the other loads are connected to the alternator. This results in a load-dump pulse, which can damage downstream circuitry. This application note provided three methods to solve load-dump issues, first by isolating the load from the source using the MAX16126, then by limiting the peak power rating using a TVS diode, and finally our recommended hybrid solution that combines the best features of both methods.

To demonstrate the robustness of the hybrid solution we sent it to an independent lab, which verified that our device and solution met the ISO 16750-2 specification, protecting 24V automotive systems against load-dump pulse events.

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